

Appx C
(Refers S No 4 (a) of S of C)

QUALITATIVE/ TECHNICAL SPECIFICATION OF ITEMS : JOB NO 2033 AND "ESTABLISHMENT OF INCIDENT MANAGEMENT CELL (IMC) AT DIV HQ COMPLEX" OUT OF SRE SCH-11 REV PROC (CODE HEAD 439/08) FOR FY 2026-27

1. AI Workstation Server

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Processor	Intel Core i9-13900K or higher			
2.	GPU	NVIDIA RTX 4090 with 24 GB VRAM or better			
3.	RAM	Minimum 128 GB DDR5 or higher			
4.	Storage	Minimum 2 TB NVMe SSD or higher			
5.	Cooling	Liquid cooling system			
6.	Motherboard	Compatible with DDR5 & PCIe Gen4/ Gen5			
7.	Power Supply	Minimum 1000W 80+ Gold or better			
8.	Networking	Min 1 x 2.5 Gigabit LAN Port			
9.	Ports	Minimum 4 USB 3.0/ HDMI/ DP or higher			
10.	OS Compatibility	Windows 11 Pro latest Version pre-loaded with DVD pack			
11.	OEM Certification	ISO 9001:2015, ISO 14001:2015, ISO 45001:2018, ISO 27001:2022, and ISO 50001:2018, SA 8000			
12.	Make in India	OEM shall comply with Government of India Make in India (MII) policy guidelines.			
13.	Warranty	3 Years			

2. IMC Management Console

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Processor	Intel Core i5 14th Gen/Ultra 5 125U or Higher			
2.	Chipset	Intel SoC Architecture			
3.	RAM	8 GB DDR5 Expendable Up 64 Gb			
4.	BIOS	BIOS of the Console to have tool for enhanced security feature like self-healing, regular checks.			
5.	Drive Controller	Integrated SATA controller, min 4 x 6Gbps ports, RAID 0,1			
6.	Storage	1TB SSD Or Higher			
7.	Optical Drive	DVD Writer			
8.	Graphics	Integrated			
9.	Keyboard/Mouse	USB Keyboard & Optical Mouse			
10.	Audio	Integrated HD Audio			

11.	Communication	LAN + WIFI 6 + Bluetooth 5.3			
12.	Expansion Slots	PCIe Gen5/Gen3 slots + M.2 support			
13.	Ports	USB, HDMI, DP, RJ45			
14.	OS	System should be supplied with pre-installed, factory loaded genuine Windows OS, Windows 11 Professional 25H2 or Latest Version with DVD Pack			
15.	Form Factor	Convertible Mini Tower			
16.	Power Supply	Min 700W, 92% efficient			
17.	Monitor	23.8-inch IPS, FHD			
18.	Warranty	3 Years On-site			
19.	Product Certification	EPEAT Gold India Registered, Energy Star, WEEE, Leading ISV's			
20.	OEM Certification	ISO 9001:2015, ISO 14001:2015, ISO 45001:2018, ISO 27001:2022, and ISO 50001:2018, SA 8000			
21.	Make in India	OEM shall comply with Government of India Make in India (MII) policy guidelines.			

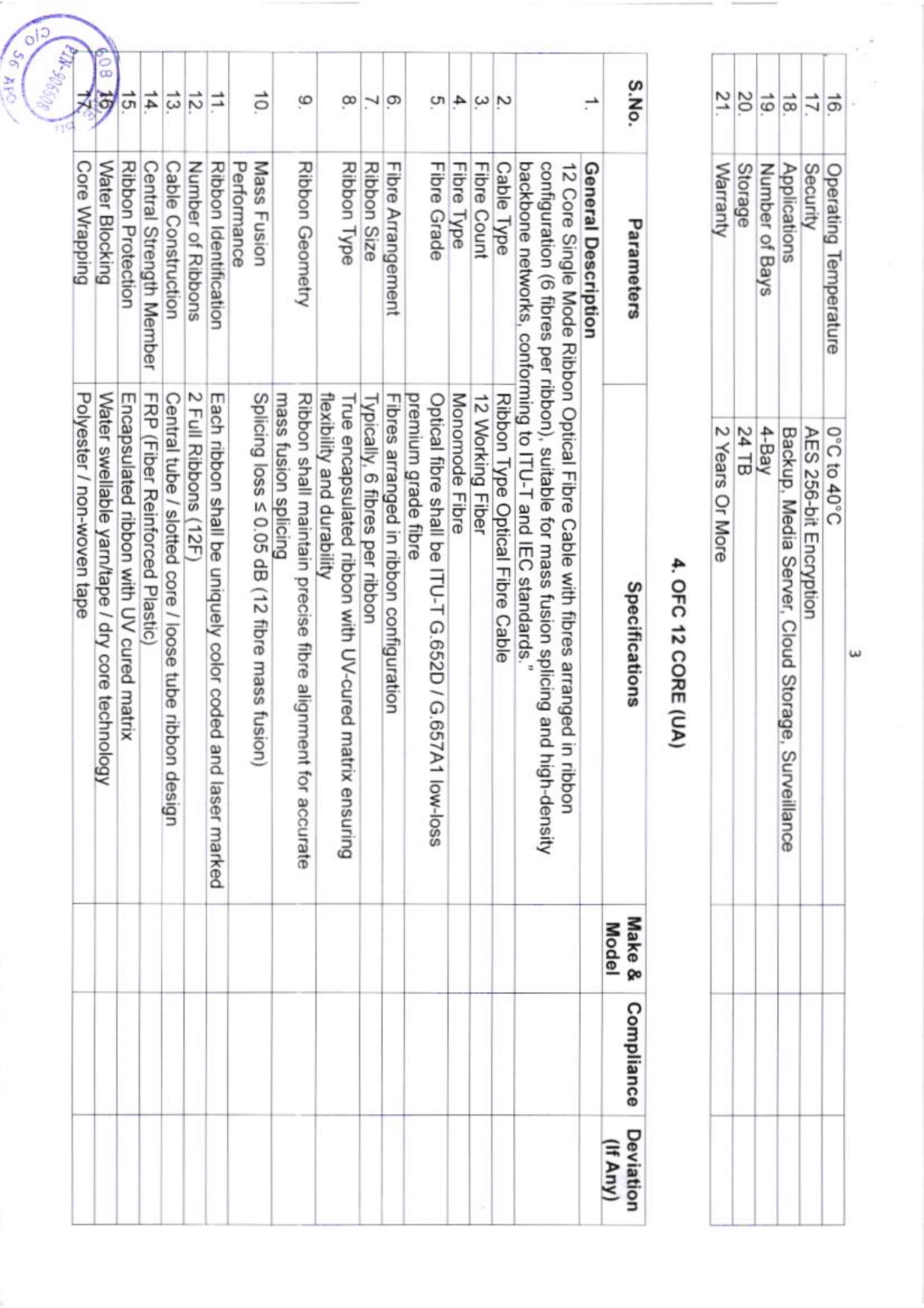
3. 24 TB NAS With HDD

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Number of Bays	4-Bay			
2.	Drive Type Supported	3.5" SATA HDD / 2.5" HDD/SSD			
3.	Max Raw Capacity	Up to 72 TB (depending on drives)			
4.	Processor	Realtek RTD1296 Quad-Core 1.4 GHz			
5.	Memory	1 GB DDR4 (non-expandable)			
6.	Ethernet Port	1 x 2.5 Gigabit Ethernet			
7.	USB Ports	2 x USB 3.2 Gen 1			
8.	RAID Support	RAID 0, 1, 5, 6, 10, JBOD			
9.	File System	EXT4 (Internal), FAT32/NTFS/Ex FAT (External)			
10.	Power Supply	External Adapter			
11.	Power Consumption	Approx. 17W (Operation)			
12.	Cooling System	120 mm Fan			
13.	Operating System	ADM (ASUSTOR Data Master)			
14.	Hot Swap	Supported			
15.	Noise Level	Approx. 19 Db			

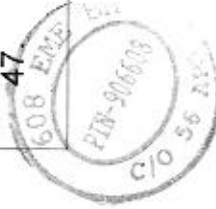
16.	Operating Temperature	0°C to 40°C			
17.	Security	AES 256-bit Encryption			
18.	Applications	Backup, Media Server, Cloud Storage, Surveillance			
19.	Number of Bays	4-Bay			
20.	Storage	24 TB			
21.	Warranty	2 Years Or More			

4. OFC 12 CORE (UA)

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	General Description				
	12 Core Single Mode Ribbon Optical Fibre Cable with fibres arranged in ribbon configuration (6 fibres per ribbon), suitable for mass fusion splicing and high-density backbone networks, conforming to ITU-T and IEC standards."				
2.	Cable Type	Ribbon Type Optical Fibre Cable			
3.	Fibre Count	12 Working Fiber			
4.	Fibre Type	Monomode Fibre			
5.	Fibre Grade	Optical fibre shall be ITU-T G.652D / G.657A1 low-loss premium grade fibre			
6.	Fibre Arrangement	Fibres arranged in ribbon configuration			
7.	Ribbon Size	Typically, 6 fibres per ribbon			
8.	Ribbon Type	True encapsulated ribbon with UV-cured matrix ensuring flexibility and durability			
9.	Ribbon Geometry	Ribbon shall maintain precise fibre alignment for accurate mass fusion splicing			
10.	Mass Fusion Performance	Splicing loss ≤ 0.05 dB (12 fibre mass fusion)			
11.	Ribbon Identification	Each ribbon shall be uniquely color coded and laser marked			
12.	Number of Ribbons	2 Full Ribbons (12F)			
13.	Cable Construction	Central tube / slotted core / loose tube ribbon design			
14.	Central Strength Member	FRP (Fiber Reinforced Plastic)			
15.	Ribbon Protection	Encapsulated ribbon with UV cured matrix			
16.	Water Blocking	Water swellable yarn/tape / dry core technology			
17.	Core Wrapping	Polyester / non-woven tape			



18.	Rip Cord	Provided		
19.	Outer Sheath	UV Stabilized HDPE / LSH		
20.	Armouring	As per requirement (Unarmoured / Armoured optional)		
21.	Attenuation (Low Loss)	≤ 0.34 dB/km @ 1310 nm and ≤ 0.21 dB/km @ 1550 nm		
22.	Operating Wavelength	1310 / 1550 / 1625 nm		
23.	Cladding Diameter	$125 \pm 1 \mu\text{m}$		
24.	Coating Diameter	$245 \pm 10 \mu\text{m}$		
25.	Tensile Strength	≥ 5000 N		
26.	Bending Radius	$\geq 15 \times$ Cable Diameter		
27.	Crush Resistance	≥ 3000 N/100 mm		
28.	Impact Resistance	As per IEC 60794		
29.	Operating Temperature	-30°C to $+70^\circ\text{C}$		
30.	Installation Temperature	-20°C to $+70^\circ\text{C}$		
31.	Storage Temperature	-30°C to $+70^\circ\text{C}$		
32.	Fibre Colour Coding	Standard 12 colour coding within ribbons		
33.	Ribbon Identification	Each ribbon uniquely colour coded or numbered		
34.	Drum Length	$2 \text{ Km} \pm 10\%$		
35.	Installation Method	Duct / Blowing / High-density network		
36.	Splicing Method	Compatible with mass fusion splicing		
37.	Compliance Standards	IEC 60794, IEC 60793, ITU-T G.652D, Telecordia GR-20		
38.	Dielectric Nature	100% Non-metallic		
39.	Marking	Sequential meter marking on sheath		
40.	Fibre Proof Test	Each fibre shall be proof tested at ≥ 100 kpsi		
41.	Cable Core Type	Dry core (gel-free) technology for easy handling and maintenance		
42.	Operating Environment	Resistant to UV Radiant, Moisture ingress, Fire and Rodent Damage		
43.	Water Blocking Efficiency	Zero water penetration over 3 meters for 24 hours		
44.	Micro-bending Loss	Extremely low micro-bending loss as per ITU-T standards		
45.	Fibre Consistency	All fibres shall have uniform attenuation and dispersion characteristics		
46.	Sheath Quality	FR sheath with anti-rodent and anti-termite properties		
47.	Interwoven Steel Wires	At Least 2 Galvanized Steel Wires, Diameter, 0.8-1.0mm, Zinc Coating : $\geq$		



48.	UV Stability	Outer sheath shall withstand long-term UV exposure without degradation			
49.	Crush Resistance	≥ 5000 N/100 mm (High strength category)			
50.	Impact Resistance	Suitable for heavy-duty outdoor installation as per IEC standards			
51.	Tensile Strength	≥ 6000 N (High tensile ribbon cable)			
52.	Temperature Performance	Stable performance from -40°C to +70°C			
53.	Installation Compatibility	Suitable for high-speed air blowing installation			
54.	Life Expectancy	Minimum operational life ≥ 25 years			
55.	Manufacturing Quality	Fully automated ribbon manufacturing process with inline quality monitoring			
56.	Marking	Meter marking, Manufacturer name, Fibre count			
57.	Type Test	Cable must be type tested from NABL accredited lab / TEC approved lab			
58.	Factory Test	100% OTDR & attenuation test report to be provided with supply			
59.	Marking	Meter marking, Manufacturer name, Fibre count			
60.	Make in India	OEM shall comply with Government of India Make in India (MII) policy guidelines.			
61.	Product Certification	UL, GMP, RoHS, FC Certification			
62.	OEM Certification	ISO 9001:2015, 14001:2015, 45001:2018, 10002:2018			
63.	R&D	OEM Must be in house R&D approved by GOI			
64.	Warranty	The offer Optical fiber cable (OFC) must come up with 3-year warranty, covering defects in material and workmanship, from the date of installation, in case of failure			

5. PTZ Camera 2 MP with Fixture

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
608 E1	Type of Camera	PTZ (Day & Night)			
2	Image Sensor	1/2.8" 2MP PS CMOS Image sensor (0.9071 centimeters)			



3.	Effective Pixels	1920 (H) × 1080 (V)			
4.	Electronic Shutter	1/1 s–1/30,000 s			
5.	Max. Aperture	F1.6–F3.5			
6.	RAM	512 MB			
7.	ROM	256 MB			
8.	Minimum Illumination	0.005 lux@F1.6 (Color, 30 IRE), 0.0005 lux@F1.6 (B/W, 30 IRE), 0 lux (IR light)			
9.	White Balance	Auto, Indoor, Outdoor, Tracking, Manual, Sodium lamp, Natural light, Street lamp			
10.	S/N Ratio	≥55 dB			
11.	Focus Control	Auto, Semi-auto, Manual			
12.	Lens	4.8 mm–120 mm			
13.	Close Focus Distance	0.1 m–1.5m			
14.	Angle of View	H: 60.6°–3.2°; V: 34.1°–1.8°; D: 69.4°–3.7°			
15.	Iris Control	Auto, Manual			
16.	DORI Distance	1655.2 mm (Detect), 653.8 mm (Observe), 331.0 mm (Recognize), 165.5 mm (Identify)			
17.	Back Light Compensation	BLC / HLC / WDR (120dB)			
18.	IR Control	Zoom Prio, Manual, Auto			
19.	IR Number	3 IR			
20.	Day/Night	Auto (ICR) / Color / B/W			
21.	IR Range	100 Mtr.			
22.	Video Format	PAL			
23.	Bit Rate	H265: 86 kbps–5632 kbps, H264: 256 kbps–8192 kbps			
24.	Overlay	Channel Title, Time Title, OSD Info, PTZ Coordinate, Zoom, North, Pattern, Temperature, Preset, Location			
25.	Gain Control	Auto, Manual			
26.	Resolution	2MP (1920 × 1080), 720p, D1, CIF			
27.	Pan/Tilt Angle	Pan: 0° to 360° endless, Tilt: –15° to +90°, auto flip 180°			
28.	Optical Zoom	25×			
29.	Digital Zoom	16×			
30.	Image Rotation	180°			
31.	Defog	Electronic			
32.	Special Features	Up to 300 Presets, 5 Pattern, 8 Tour, 5 Scan			



33.	Frame Rate	Main stream: 2MP@1-25fps, 720p, D1 Sub streams supported			
34.	Bit Rate Control	CBR/VBR			
35.	IMD	Support			
36.	Noise Reduction	2D NR / 3D NR			
37.	Stream Capability	Penta stream			
38.	Idle Motion	Pattern, Preset, Scan, Tour			
39.	Protocol	IPv4, IPv6, HTTP, HTTPS, TCP, UDP, RTP, RTSP, RTCP, RTMP, SMTP V3, SFTP, DHCP, DNS, QoS, UPnP, NTP, Multicast, ICMP, IGMP, NAS, PPPoE, SNMP, TLS v1.2/v1.3, ONVIF			
40.	Device Search Tool	IP finder			
41.	OTA (Over-The-Air)	Support			
42.	SDK and API	Support			
43.	Power Off Memory	Support			
44.	Manual Speed	Pan: 0.1°-80°/s, Tilt: 0.1°-80°/s			
45.	Preset Speed	Pan: 80°/s, Tilt: 80°/s			
46.	Event Alarm	Motion Detection, Defocus Detection, Disk Full, Disk Error, Video Tampering, External Alarm, Security Warning, Audio Detection, AI Event			
47.	Alarm Linkage	Capture, Recording, Send email, Preset, Tour, Pattern, Alarm digital output			
48.	Privacy Mask	Support			
49.	IVS (Perimeter Protection)	Tripwire, Intrusion, Parking Detection, Crowd Gathering, Missing Object, Loitering Detection, Face Detection, IMD			
50.	Cyber Security	AES 256-bit Encryption, Configuration encryption, Trusted execution, Digest, Security logs, HTTPS, Firmware encryption			
51.	Audio In	1 Channel			
52.	Audio Out	1 Channel			
53.	Alarm IN	2 (0-5 VDC)			
54.	Alarm Out	1 Channel			
55.	Reset Button	Support			
56.	Weatherproof Standard	IP66, TVS 6000V lightning proof, Surge protection			
57.	Storage	SFTP, NAS, Micro SD card (Max. 512GB)			
58.	Power Source	12 VDC, 3A ±10%			



59.	Power Consumption	Basic: 6.5W, Max: 17.25W			
60.	Operating Temperature	-40°C to +65°C			
61.	OEM Certificates	MIL-STD-883C, ISO 14001, ISO 45001, CE, FCC, RoHS, BIS, STQC, approved			
62.	Product Certification	OEM/Bidders offered products which have STQC (Standardization Testing & Quality Certification ER cert compliance as per MEITY notification regarding PPO dt 06 Mar 2024).			
63.	Warranty	2 Years or More			

6. Optical to Ethernet Converter (1G)

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Item Description	Optical Fibre Joint Enclosure			
2.	Capacity	Suitable for 12 fibre cores			
3.	Material	High quality ABS / Engineering plastic body			
4.	Protection Rating	IP66 (waterproof & dustproof)			
5.	Splice Tray	Modular splice trays for fibre management			
6.	Cable Entry Ports	Multiple ports with proper sealing arrangement			
7.	Compatibility	Supports single mode and multi-mode fibres			
8.	Operating Temperature	-40°C to +70°C			
9.	UV Resistance	UV stabilized for outdoor use			
10.	Mechanical Strength	Resistant to shock, vibration and environmental stress			
11.	Sealing	Rubber gasket sealing for complete waterproofing			
12.	Locking Mechanism	Secure locking to prevent unauthorized access			
13.	Splice Protection	Compatible with heat shrink splice protection sleeves			
14.	Mounting	Suitable for underground / aerial / wall mounting			
15.	Certification	IP66 certified and TEC compliant, UL94-V0			
16.	Standards	IEC / RoHS / Telecom standards compliant			
17.	Fire Resistance	Compliant with UL94-V0 Fire Rating to Ensure Minimal Fire Hazard in Case of Emergencies			
18.	Compliance Standards	Meets RoHS standard for Environmental protection and safety			
19.	Warranty	1 Year or More			



9
7. NVR 128 Channel

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Processor	Industrial-grade processor			
2.	Operating System	Embedded Linux			
3.	IP Camera Input	128 Channels			
4.	User Interface	Web, Local GUI			
5.	AI by Camera	Face detection, Face recognition, Video metadata, Perimeter protection, IMD, Stereo analysis, Crowd distribution, People counting, ANPR, Heat map, PPE detection			
6.	AI by NVR	Face detection, Face recognition, Video metadata, Perimeter protection, IMD			
7.	Perimeter Performance AI by NVR	24 Channels, 10 IVS rules for each channel			
8.	Perimeter Performance AI by Camera	Human and vehicle detection on all channels, supporting up to 32 targets/sec			
9.	Video Output	2 VGA: 1920×1080, 1280×1024, 1280×720 2 HDMI: 3840×2160, 1920×1080, 1280×1024, 1280×720			
10.	Face Detection AI by NVR	4 or 8 channels depending on 4K mode			
11.	Face Detection AI by Camera	32 Channels			
12.	Face Recognition AI by NVR	Supports FD + FR by recorder/camera in multiple modes			
13.	Face Recognition AI by Camera	32 Channels			
14.	Face Attributes	6 Attributes			
15.	Face Database Capacity	Up to 20 databases with 200,000 images, total 44G			
16.	Decoding Capability	Supports up to 32MP decoding with AI enabled/disabled configurations			
17.	Intelli Check AI by Camera + NVR	Max. 36 Channels			
18.	License Plate Database Capacity	Up to 20,000 plate numbers, blacklist and allowlist			
	Motor Vehicle Attributes	9 Attributes			



20.	Non-motor Vehicle	4 Attributes			
21	Disk Interface	8 SATA ports, up to 20TB each			
22	Abnormality Alarm	Disk error, low space, RAID exception, network disconnection, IP conflict, video loss, tampering, camera offline, audio detection			
23	IMD+ by NVR	32 Channels with secondary filtering for humans and vehicles			
24	IMD+ by Camera	Human and vehicle detection on all channels, supporting up to 64 targets/sec			
25	ANPR by Camera	Human and vehicle detection on all channels, supporting up to 32 targets/sec			
26	Metadata Performance AI by NVR	8 Channels, 23 attributes extraction			
27	Metadata Performance AI by Camera	32 Channels			
28	Multi Screen Preview	Main screen: 1/4/8/9/16/25/36/64; Sub screen: 1/4/8/9/16			
29	Network Mode	Multi-address, Load balance, Fault tolerance			
30	Network Bandwidth	512 Mbps incoming/recording/outgoing (AI disabled) 200 Mbps incoming/recording/outgoing (AI enabled)			
31	Resolution	32MP, 24MP, 16MP, 12MP, 8MP, 6MP, 5MP, 4MP, 3MP, 2MP, 960p, 720p, D1, CIF, QCIF			
32	People Attributes	9 Human body attributes			
33	General Alarm	Motion detection, local alarm, network alarm, PIR alarm, thermal alarm			
34	Intelligent Alarm	Face detection, face recognition, video metadata, perimeter protection, license plate comparison			
35	RAID	RAID 0/1/5/6/10			
36	Alarm Linkage	Record, snapshot, local alarm output, IPC alarm output, buzzer, email			
37	Recording Mode	General, Motion detection, Intelligent, Alarm, POS			
38	Playback Mode	Instant playback, event playback, smart playback			
39	Multi-Channel Playback	Up to 16 channels			
40	Protocols	HTTP, HTTPS, TCP/IP, IPv4, IPv6, UDP, DHCP, DNS, SMTP, ONVIF, CGI, SDK			
41	Power Redundancy	Support			
42	Video Compression	Instastream, H.265, H.264+, H.264, MJPEG			
43	Disk Group	Support			



44	Hot Swapping	Support			
45	Audio Compression	G.711a, G.711u, PCM, G726			
46	Audio Input	1 Channel RCA			
47	Audio Output	2 Channel RCA			
48	Alarm Input	16 Channels			
49	Alarm Output	8 Channels			
50	Backup Mode	USB device and network			
51	Interface Ports	4 USB, 2 HDMI, 2 VGA, 2 RS485, 1 RS232, 1 eSATA			
52	Ethernet Port	2 × 10/100/1000 Mbps RJ-45			
53	Smart Phone	iOS, Android			
54	Browser	Chrome, IE, Safari, Edge, Firefox			
55	Power Supply	100–240 VAC, 50/60 Hz			
56	Power Consumption	<13W (without HDD)			
57	Operating Temperature	–10°C to +55°C			
58	Operating Humidity	10%–93% RH			
59	Storage Temperature	–20°C to +60°C			
60	OEM Certificates	MIL, ISO 9001, ISO 14001, ISO 45001, CE, FCC, RoHS, BIS, STQC, approved			
61	Product Certification	OEM/Bidders offered products which have STQC (Standardization Testing & Quality Certification ER cert compliance as per METTY notification regarding PPO dt 06 Mar 2024).			
62	Warranty	2 Years or More			

8. 24 Port L2 Manageable POE Switch

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Switch Type	Layer 2 Switch			
2.	Ports	24 × 10/100/1000 Mbps RJ45 (Auto MDI/MDIX), 4 × 1G Base-T/SFP Combo Ports			
3.	Standards	IEEE 802.3, 802.3u, 802.3ab, 802.3z, 802.3az, 802.3x, 802.3af/at			
4.	Switching Capacity	Minimum 56 Gbps			
5.	Forwarding Rate	41.7 MPPS			
6.	MAC Table	Minimum 8K MAC entries			

7.	Buffer Memory	4.1 Mbits		
8.	Temperature	Operating: -5°C to 50°C, Storage: -20°C to 70°C		
9.	Loop Detection	Loop detection from unmanaged network		
10.	Multicast	IGMP v1/v2, MLD v1/v2 (256 groups), Jumbo Frame up to 10,000 bytes		
11.	Discovery	LLDP, LLDP-MED		
12.	Power Saving	Link status saving, hibernation, port shutoff, cable length detection		
13.	VLAN	4K VLAN IDs, 256 static VLANs, Port/Voice/Surveillance VLAN		
14.	Port Mirroring	One-to-One / Many-to-One		
15.	QoS	802.1p, WRR, Strict, Bandwidth Control, ToS		
16.	Routing	4 IP Interfaces, Static & Default Routing, 100 IPv4 & 50 IPv6 routes		
17.	ACL	Port, MAC, IP based, L2 & L3 ACL (IPv4/IPv6)		
18.	Security	DoS prevention, 802.1X, Port Security, ARP spoofing prevention, DHCP screening, IP-MAC binding		
19.	Spanning Tree	STP, RSTP, MSTP with loop protection and CPU safeguard		
20.	Authentication	802.1X, RADIUS, EAP, OTP, TLS, TTLS, PEAP		
21.	Features	Cable diagnostics, IP inspection, SSH v2, TLS v1.0, Flow Control		
22.	Management	Web GUI, CLI, Dual image, Dual configuration		
23.	Monitoring	SNMP v1/v2c/v3, RMON, Syslog, SNMP		
24.	Green Tech	System hibernation, Time-based PoE, Surveillance mode (ONVIF/NVR)		
25.	Certifications	CB, UL, ROHS, MTC, IEC 62443-4-1, STQC		
26.	OEM Compliance	ISO 45001, ISO 27001, ISO 14001, IEC 62443-4-1		
27.	Warranty	2 Years or More		

9. 8 Port L2 Manageable POE Switch

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Network Port	PoE (1~6): 10/100/1000Mbps PoE ports (RJ45) PoE (7~8): 10/100/1000Mbps PoE ports, BT port LAN (1~2): 10/100/1000Mbps uplink port (RJ45) SFP (1~2): 10/100/1000Mbps uplink ports (SFP)			



2.	Network Standard	IEEE 802.3i 10BASE-T IEEE 802.3u 100BASE-TX / IEEE 802.3ab 1000BASE-T IEEE 802.3x Flow Control IEEE 802.3af/at/bt Power over Ethernet			
3.	Transmission Rate	2000Mbps Full Duplex; 10/100/1000Mbps Half Duplex			
4.	Transmission Distance	1~100m			
5.	MAC Address Table	8K			
6.	Bandwidth	24.0 Gbps			
7.	Network Latency	≤16μs (10 to 100Mbps, 64-byte package size)			
8.	Frame Filter & Transmission Rate	14800pps for 10Mbps port 148800pps for 100Mbps port 148800pps for 1000Mbps port			
9.	Power Budget	120W			
10.	Power Input	AC 180~240V, Total ≤120W			
11.	POE Watchdog	Port 1~8			
12.	Power Output	POE 1~6 Port: 1/2(+52V), 3/6(GND) POE 7~8 Port: 1245(+53V), 3678(GND), compatible 1/2(+52V), 3/6(GND) ≤60W			
13.	Lightning Protection	6KV			
14.	Operating Temperature	-30°C ~ 60°C			
15.	Working Humidity	10%~90% (Non-condensing)			
16.	Storage Temperature	-40°C ~ 70°C			
17.	Storage Humidity	5%~90% (non-condensing)			
18.	Dimensions	187mm(L) x 139mm(W) x 43mm(H)			
19.					
20.	Package Content	AC Power Cable, Installation Manual			
21.	Certifications	ISO 45001, ISO 27001, ISO 9001, ISO 14001, ISO 14064-1			
22.	Cyber Security Compliance	IEC 62443-4-1:2018 or equivalent			
23.	Government Compliance	Trusted Telecom Portal (DoT, India)			
24.	Warranty	2 Years or More			



14
10. SFP Module 20 Km

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Product Type	SFP Optical Transceiver Module			
2.	Form Factor	SFP (Small Form-factor Pluggable)			
3.	Data Rate	1.25 Gbps (Gigabit)			
4.	Transmission Distance	Up to 20 KM			
5.	Fiber Type	Single Mode (SMF)			
6.	Connector Type	LC Connector			
7.	Wavelength	1310 nm (Tx/Rx)			
8.	Interface	LC Duplex			
9.	Compatibility	Compatible with standard SFP ports			
10.	DDM Support	Digital Diagnostic Monitoring (Optional)			
11.	Power Supply	3.3V			
12.	Power Consumption	≤ 1W			
13.	Operating Temperature	0°C to +70°C			
14.	Storage Temperature	-40°C to +85°C			
15.	Hot Pluggable	Yes			
16.	Standards	IEEE 802.3z, SFP MSA Compliant			
17.	Application	Fiber Networking, ISP, Data Centers, CCTV			
18.	Transmission Distance	Up to 20 KM			
19.	Fiber Type	Single Mode (SMF)			
20.	Warranty	1 Year			

11. Optical to Ethernet (OTE) Assembler

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Product Type	Fiber Media Converter (Optical to Ethernet)			
2.	Interface Type	RJ45 (Ethernet) to Fiber (SC Port / SFP variant)			
3.	Ethernet Port	1 × 10/100/1000 Mbps RJ45			
4.	Fiber Port	1 × SC (Single Mode) / SFP Slot (Model dependent)			
5.	Supported Speed	10/100/1000 Mbps Auto-Negotiation			



6.	Transmission Distance	Up to 20 KM (Single Mode Fiber)			
7.	Wavelength	1310 nm / 1550 nm (Model dependent)			
8.	Fiber Type	Single Mode (SMF)			
9.	Duplex Mode	Full / Half Duplex Auto-Sensing			
10.	Auto MDI/MDI-X	Supported			
11.	Transmission Method	Store-and-Forward / Cut-Through			
12.	Jumbo Frame	Up to 9K Supported			
13.	Packet Size	Up to 1518 Bytes			
14.	Network Features	VLAN / QoS / VoIP Support			
15.	Flow Control	Supported (IEEE 802.3x)			
16.	STP Support	Yes			
17.	LED Indicators	Power, Link/Activity, Speed Status			
18.	Power Supply	External Adapter (AC 220V)			
19.	Power Consumption	≤5W (approx.)			
20.	Mounting	Desktop / Wall Mount / DIN Rail			
21.	Operating Temperature	0°C to +50°C			
22.	Application	CCTV, ISP Network, Long Distance LAN, FTTH			

12. Digital Display Device 55 Inch

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Light Source	LED backlight			
2.	Resolution	3840 × 2160 (4K UHD) or better			
3.	Refresh Rate	60Hz or better			
4.	Aspect Ratio	16:9 or better			
5.	Display Colour	1.07 Billion colors (10-bit) or higher			
6.	Surface	4mm Anti-glare tempered glass			
7.	Brightness	Upto 450 cd/m ² or better			
8.	Haze	25%			
9.	Contrast Ratio	1200:1 or better			
10.	Viewing Angle	178° (H/V)			
11.	Backlight Life	≥ 50,000 hours or better			
13.	Touch Technology	IR Touch			
13.	Touch Points	40 touch points on Android & Windows or better			

14.	Touch Response	≤ 5 ms or less			
15.	Touch Accuracy	± 2 mm or less			
16.	CPU	Octa-Core A55 or better			
17.	GPU	Quad-Core MaliG52 or better			
18.	RAM + ROM	8GB RAM + 128GB Storage or better			
19.	Operating System	Android 14.0 or better			
20.	Wi-Fi	802.11 a/b/g/n/ac (2.4GHz/5GHz)			
21.	Bluetooth	5.0 or latest			
22.	Speakers	2 x 20 W or better			
23.	Interfaces	4x USB 2.0, 1x USB 3.0, 1x Type-C, 1x TF Card Slot, 3x HDMI IN (1 ARC), 1x Touch USB, 1x RS232, 1x Line Out, 1x Coax, 1x RJ45 IN, 1x 3.5mm Jack, 1x VGA IN, 1x VGA Audio IN			
24.	Whiteboard Features	Multi-touch interactive whiteboard (upto 20 points), handwriting-to-text, AI shape recognition, math/science tools, sticky notes, text editing, real-time collaboration, export to PDF, YouTube & Google integration, browser panel, 3D model viewer, screen capture			
25.	Accessories	Power Cord x1, Stylus Pen x2, Wall Mount Kit x1, Remote Control x1, User Manual x1			
26.	Certifications	BIS, CE, FCC & RoHS, MII			
27.	Warranty	3 Years			

13. HDMI Matrix Switcher 8x8

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Power Source	100-240V AC, 50/60Hz			
2.	Operating Temperature	0° to +40°C			
3.	Storage Temperature	-10° to +60°C			
4.	Humidity	10% to 90%, RH (non-condensing)			
5.	Product Dimensions	19" rack mountable			
6.	Video Input	08 x HDMI Type A			
7.	Video Output	08 x HDMI Type A			



8.	Compliance	HDMI (3D, Deep Colour) HDCP 1.4 compatible		
9.	Resolution	Up to 1080p		
10.	Audio Output	3.5 mm Mini Stereo Jack Female		
11.	Control	1 x DB-9 Female (Black)		
12.	Ethernet	Minimum 1 x RJ-45 Female		
13.	Switch Power	Minimum 1 x Rocker Switch		
14.	Connectors Power	1 x 3-Prong AC Socket		
15.	Maximum Input Power Rating	100-240 VAC; 50-60 Hz; 1.0A		
16.	Array Mode Feature	Yes, with Array port		
17.	Warranty	1 Year		
18.	Additional Features			
19.	Should be complete with rack mounting kit & power cord. Should be Audio-enabled HDMI audio can be extracted to stereo audio, should support multiple options for configurations including front-panel pushbuttons, RS-232 control, and Ethernet connections for web-based GUI. Should support scaling feature in order to convert input resolutions to the optimum display resolutions. Should have a Local HDMI output for preview of 16 source, should have features such as seamless switching in order to avoid delay in video streams, ESD protection for HDMI, Supports Dolby True HD and DTS HD Master audio, prevents image tearing by synchronizing the output frame rate to the input signal frame rate, should have video scaling function to convert input resolutions to the optimum display resolutions.			
20.	Make in India Compliance	OEM shall comply with Government of India Make in India (MII) policy guidelines.		

14. 9U Rack Wall Mount

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Rack Size	9U			
2.	Standard	DIN 41494 or equivalent			
3.	Construction	Welded or CKD (Knock Down)			
4.	Front Door	Lockable Toughened Glass Door			
5.	Frame Type	Mild Steel			
6.	Equipment Mounting	DIN Standard 19 inch slots			
7.	Mounting Angle	19 inch mounting profile			

8.	Finish	Powder Coated		
9.	Top & Bottom Cover	Vented with cable entry provision		
10.	Cooling	Provision for 1 or 2 cooling fans		
11.	Side Panels	Removable		
12.	Color	Grey / Off White / Black (RAL 7035 / RAL 9005)		
13.	Load Capacity	Up to 30 Kg		
14.	Height	Approx. 450 mm		
15.	Width	Approx. 550 mm		
16.	Depth Options	400 / 500 / 600 mm		

15. UPS 5 KVA with 60 Min Backup

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Rating in KVA	5.0 KVA			
2.	Switching Technology	IGBT-PWM			
3.	Input Voltage	AC Single Phase 160V - 260V, 50Hz			
4.	Output Voltage	AC Single Phase 230V $\pm 1\%$, 50Hz			
5.	Warranty for UPS	Minimum 3 Years or More			
6.	Battery / Backup Bank	UPS shall be supplied with Battery Bank and comply with latest GST Council recommendations.			
7.	Type of Battery	SMF-VRLA			
	Backup Time	60 Minutes or Higher			
	Minimum Guaranteed VAH	1600 $\times$ Per KVA $\times$ Per Hour of Backup Time			
8.	Warranty for Battery	Minimum 2 Years or More			
9.	Accessories	Battery Rack/Trolley, Cabling 6 meters (Input up to 3m and Output up to 3m), Parallel Kit (Hot Standby Configuration), Connectors and all required accessories.			
10.	Degree of Protection	IP20 or IP21			
11.	Intended Applications	Suitable for Data Centres, Network Operation Centres, Security Operation Centres, State & District Centres, Command & Control Centres, Digital India Projects and Critical Infrastructure.			
12.	Type of Design / Construction	Floor Standing			



13.	Mandatory Certifications & Compliance	BIS Certified, NABL/ILAC Accredited Test Report or Equivalent, Make in India (MII), CE Certified, RoHS Compliant, IEC 62040-1, IEC 62040-2 and IEC 62040-3 Compliance mandatory.			
14.	OEM Certifications	OEM/Bidder should have valid ISO 9001:2015, ISO 14001:2015 and ISO 45001:2018 certifications. ISO 27001 and ISO 50001 preferred.			
15.	Battery Certifications	Battery should comply with IEC 60896 standards. In case of Lithium Battery Bank, IEC 62619, IEC 62133 and UN38.3 certifications mandatory.			
16.	Protection Features	Short Circuit Protection, Overload Protection, Surge Protection, Lightning Protection, Battery Deep Discharge Protection, Auto Restart, Static Bypass and Emergency Power Off (EPO).			
17.	Performance Requirements	THDi <3%, Input Power Factor >0.99 and Efficiency $\geq 95\%$.			
18.	Monitoring & Communication	SNMP / Remote Monitoring and MODBUS support preferred.			
19.	Mandatory Certificates The offered UPS System with Battery Bank shall be BIS certified and compliant with IEC 62040-1, IEC 62040-2 and IEC 62040-3 standards. The bidder/OEM shall submit valid NABL/ILAC accredited type test reports along with ISO 9001:2015, ISO 14001:2015 and ISO 45001:2018 certifications. The UPS system shall be CE certified and RoHS compliant. Battery system shall comply with IEC 60896 standards and in case of Lithium battery bank, IEC 62619, IEC 62133 and UN38.3 certifications shall be mandatory				
20.	Mandatory Test Reports Valid NABL/ILAC accredited laboratory type test reports for the offered UPS System and Battery Bank shall be mandatory. Test reports must cover electrical safety, performance, efficiency, overload capability, harmonic distortion, environmental compliance and protection parameters as per applicable IEC standards.				

16. UPS 1 KVA with 30 Min Backup

S. No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1	Topology	Line Interactive			



2.	Input Voltage	220/230/240 VAC			
3.	Input Voltage Range	140 – 300 VAC			
4.	Input Frequency	40–70 Hz			
5.	Output Capacity	600VA / 1000VA			
6.	Output Power	360W / 600W			
7.	Waveform	Simulated Sine Wave			
8.	AVR	Automatic Voltage Regulation (Double Boost/Single Buck)			
9.	Transfer Time	4 ms Typical			
10.	Battery Type	Sealed Lead Acid			
11.	Recharge Time	Approx. 8 Hours			
12.	LED Indicators	Power On, Battery Mode			
13.	Audible Alarm	On Battery, Low Battery, Overload, Fault			
14.	Form Factor	Brick Type			
15.	Operating Temperature	0°C to 40°C			
16.	Humidity	0–90% non-condensing			
17.	Output Waveform	Simulated Sine Wave			
18.	Output Protection	Surge Protected Outlets			
19.	Fire Safety	Fire Retardant Design			
20.	Protection Features	Overload Protection			
21.	Noise Control	Silent Mode (Optional)			
22.	Mounting Type	Wall Mounting (Optional)			
23.	Form Factor	Brick Type Design			
24.	Low Load Handling	No Load Auto Shutdown Disabled			
25.	Service Support	Onsite Service Support			
26.	Certifications	BIS, NABL Test Report or Equivalent, MII			
27.	Warranty	3 Years			

17. Power Strip 05 Socket

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Number of Sockets	5 Universal Sockets			
2.	Switch Type	1 Master On/Off Switch			
3.	Rated Voltage	220–240V AC			
4.	Rated Current	6A / 10A (Model dependent)			



5.	Max Load	Up to 1500W / 2500W			
6.	Socket Type	Universal / Indian Standard			
7.	Protection	Overload Protection			
8.	Cable Length	1.5m / 3m (Optional)			
9.	Cable Type	Copper Wire, PVC Insulated			
10.	Plug Type	3 Pin India Plug			
11.	Body Material	Fire Retardant ABS Plastic			
12.	Indicator	LED Power Indicator			
13.	Mounting Type	Wall Mountable / Desktop			
14.	Safety Features	Child Safety Shutter			

18. Passive BNC DC to RJ45 Adapter

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Connector Type	BNC + DC to RJ45			
2.	Video Input	1 × BNC Female			
3.	Power Connector	1 × DC Female (2.1mm)			
4.	Output	1 × RJ45 Port			
5.	Transmission Type	Passive (No Power Required)			
6.	Cable Compatibility	Cat5/Cat5e/Cat6 UTP Cable			
7.	Transmission Distance	Up to 300 meters (depending on cable)			
8.	Signal Type	Analog Video (CCTV)			
9.	Impedance	75 Ohm (BNC)			
10.	Material	ABS Plastic Housing			
11.	Installation	Plug & Play			
13.	Application	CCTV Camera Installation, DVR Connectivity			
14.	Operating Temperature	0°C to +55°C			

19. UTP Cable Cat 6

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Cable Type	Category 6 UTP (Unshielded Twisted Pair)			
2.	Application	Suitable for Indoor & Outdoor Installation			



3.	Conductor Material	100% Solid Bare Copper		
4.	Conductor Size	23 AWG (0.57 mm)		
5.	Number of Pairs	4 Twisted Pairs		
6.	Insulation	High Density Polyethylene (HDPE)		
7.	Outer Jacket	UV Resistant PVC / FRLS / LSZH (Outdoor Grade)		
8.	Bandwidth	Minimum 250 MHz		
9.	Data Rate	Up to 1 Gbps (10/100/1000 Mbps)		
10.	Operating Temperature	-20°C to +75°C		
11.	Voltage Rating	300V		
12.	Flame Retardant	FR / FRLS / LSZH Compliant		
13.	UV Protection	UV Stabilized Jacket for Outdoor Use		
14.	Moisture Resistance	Suitable for humid & outdoor conditions		
15.	Standards	ANSI/TIA-568-C.2, ISO/IEC 11801, IEC 61156		
16.	Fire Standards	IEC 60332-1, IEC 60332-3		
17.	Smoke Standards	IEC 61034 (Low Smoke)		
18.	Halogen Free	IEC 60754 (LSZH)		
19.	EMI/EMC Compliance	IEC 61000 Series		
20.	Certification	RoHS, CE, UL Listed, ISO 9001, ISO 14001		
21.	Cable Length	305 Meter Pull Box		
22.	Color	Grey / Blue / Black (Outdoor)		
23.	Marking	Meter marking on cable		
24.	Testing	100% Factory Tested & Fluke Test Compatible		
25.	Cable Type	Category 6 UTP (Unshielded Twisted Pair)		
26.	Application	Suitable for Indoor & Outdoor Installation		
27.	Make in India Compliance	OEM shall comply with Government of India Make in India (MII) policy guidelines.		

20. RJ45 Connector Cat 6

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Parameter	Specification			
2.	Category	Cat 6			
3.	Connector Type	8P8C (8 Position 8 Contact)			
4.	Cable Compatibility	Cat6 UTP Solid/Stranded Cable			
5.	Contact Material	Phosphor Bronze with Gold Plating			



6.	Gold Plating	3-50 Micron (Model dependent)			
7.	Housing Material	Polycarbonate (PC)			
8.	Color	Transparent			
9.	Termination Method	Crimping			
10.	Standards	TIA/EIA 568-B.2, ISO/IEC 11801			
11.	Operating Temperature	-10°C to +60°C			
12.	Application	LAN Networking, CCTV, Data Communication			

21. Crimping Tool

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Application	RJ45 Connector Crimping			
2.	Compatibility	RJ45 (8P8C), RJ11/RJ12 (Model dependent)			
3.	Material	High Carbon Steel			
4.	Handle Type	Ergonomic Anti-Slip Grip			
5.	Crimping Type	Manual			
6.	Functions	Crimping, Cutting, Stripping			
7.	Blade Material	Hardened Steel			
8.	Lock Mechanism	Safety Lock Available			
9.	Durability	Heavy Duty Industrial Use			
10.	Cable Support	Cat5/Cat5e/Cat6 UTP Cable			
11.	Operation	Easy Hand Operation			

22. Patch cord LC-LC 5 Mtr

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Item Description	Optical Fibre Patch Cord LC to LC			
2.	Fibre Type	Single Mode (G.652D / G.657A1)			
3.	Length	5 Meter			
4.	Connector Type	LC to LC, factory terminated			
5.	Insertion Loss	≤ 0.30 dB			
6.	Return Loss	≥ 60 dB			
7.	Jacket	LSZH, flexible and durable			
8.	Application	Fibre interconnection and patching			



23. Patch cord LC-LC 5 Mtr

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Item Description	Optical Fibre Patch Cord LC to LC			
2.	Fibre Type	Single Mode (G.652D / G.657A1)			
3.	Length	5 Meter			
4.	Connector Type	LC to LC, factory terminated			
5.	Insertion Loss	≤ 0.30 dB			
6.	Return Loss	≥ 60 dB			
7.	Jacket	LSZH, flexible and durable			
8.	Application	Fibre interconnection and patching			

24. HDMI Cable 10 Meter

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Cable Length	10 Meter			
2.	Cable Type	High Speed HDMI Cable			
3.	Version	HDMI 2.0 or Higher			
4.	Resolution Support	Up to 4K @ 60Hz			
5.	Bandwidth	18 Gbps or higher			
6.	Connector Type	HDMI Type A Male to Male			
7.	Conductor	Copper / Tinned Copper			
8.	Shielding	Triple Shielded			
9.	Jacket	PVC / Braided			
10.	Compatibility	HDCP, 3D, ARC, Ethernet			
11.	Application	Projector, LED, VC Setup			
12.	Certification	CE, RoHS			
13.	Make in India Compliance	OEM shall comply with Government of India Make in India (MII) policy guidelines.			



25. USB Extension Cable

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Cable Length	15 Meter			
2.	Connector Type	USB Type-A Male to USB Type-A Female			
3.	USB Standard	USB 2.0 (Backward Compatible with USB 1.1)			
4.	Data Transfer Speed	Up to 480 Mbps			
5.	Cable Type	Active / Repeater Cable (Recommended for 15m length)			
6.	Power Supply	Bus Powered / Optional External Power Support			
7.	Conductor Material	Copper (High Quality)			
8.	Jacket Material	PVC Insulated			
9.	Shielding	Aluminum Foil + Braided Shield			
10.	Compatibility	PC, Laptop, Printer, Camera, Keyboard, Mouse			
11.	Plug & Play	Yes			
12.	Signal Boosting	In-built Signal Repeater			
13.	Operating Voltage	5V DC			
14.	Operating Temperature	0°C to 50°C			
15.	Cable Length	15 Meter			
16.	Warranty	1 Year or More			

26. USB to Ethernet Converter

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Specification	USB 3.0 to RJ45 LAN Adapter Supports 10/100/1000 Mbps Plug & Play, Compatible with Windows/Mac			
2.	Warranty	1 Year or More			

27. Zip Tie 6 Inch

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
	Specification	Length: 6 Inch (150mm) Material: Nylon (High Strength) Self-locking, UV Resistant			

28. Flexible Conduit Pipe 2 Inch

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Specification	Size: 2 Inch Diameter Material: PVC Flexible Used for Cable Protection			

29. U Clip Plastic 100 mm

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Specification	Size: 100 mm Material: Durable Plastic Used for Pipe/Cable Holding			

30. Screw 2 Inch

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Specification	Length: 2 Inch Material: Steel (Zinc Coated) Rust Resistant, High Strength			

31. Insulation Tape 1 Inch

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Specification	Width: 1 Inch Material: PVC Electrical Tape Flame Retardant, High Insulation			

32. Cello Tape 2 Inch

S.No.	Parameters	Specifications	Make & Model	Compliance	Deviation (If Any)
1.	Specification	Standard Width (Approx 1 Inch) Transparent Adhesive Tape Strong Adhesion, General Use			

